

LABORATORY OBSERVATION / RECORD

Course:Full Stack Web Development

Course Code:23CM4121

Experiment No.:1 (task-2)

Student Name : Kottana Roshini

Roll No.:A24126552263

1. TITLE

Style the College Information Portal Web Page Using CSS3.

2. AIM

To style the previously developed College Information Portal HTML5 page using CSS3 — applying colors, fonts, borders, gradients, box-shadow, text-shadow, border-radius and hover effects — through an external, linked stylesheet in a black-and-white theme.

3. OBJECTIVE

- To understand CSS syntax and the different types of CSS selectors.
- To apply background colors and linear gradients.
- To style fonts, borders, margins and padding.
- To apply box-shadow and text-shadow effects.
- To create rounded corners using border-radius.
- To create hover effects for navigation links, buttons and content cards.
- To link an external CSS file to an HTML document.

4. THEORY

CSS3 (Cascading Style Sheets, level 3) is used to control the presentation of HTML elements, separating content (HTML) from design (CSS). CSS rules consist of a selector and a declaration block containing property-value pairs.

The CSS box model — margin, border, padding and content — determines the spacing and sizing of every element. Additional CSS3 features used here include linear-gradient() for smooth colour transitions, box-shadow and text-shadow for depth effects, border-radius for rounded corners, and the :hover pseudo-class for interactive feedback when the mouse pointer is over an element.

An external CSS file (style.css) is linked to the HTML document using <link rel="stylesheet" href="style.css"> inside the <head> section. This keeps the stylesheet reusable and separate

from the page structure, so the same design can be applied consistently across multiple pages.

5. ALGORITHM / PROCEDURE

- Create a new file named style.css in the same folder as index.html.
- Link style.css to index.html using the <link> tag inside <head>.
- Define base styles for body: font family, text colour and background colour.
- Style the header with a linear-gradient background and text-shadow.
- Style the navigation bar and add a :hover effect on navigation links.
- Style the table with borders, header shading and alternating row colours.
- Style the form with border, border-radius, padding and box-shadow.
- Add a :hover effect on the submit/reset buttons.
- Style the canvas element with a border to visually separate it from the page.
- Style the content cards using a gradient background, border-radius and box-shadow, with a :hover effect that lifts the card and deepens the shadow.
- Style the footer with a solid background colour and centred text.
- Save the file, refresh index.html in the browser and verify every styled section.

6. PROGRAM

File: style.css

```
/* =====  
Observation Task 2 — CSS3 Styling (Black & White Theme)  
===== */  
  
/* ----- CSS Syntax / Selectors / Base ----- */  
* {  
    box-sizing: border-box;  
}  
  
body {  
    margin: 0;  
    padding: 0;  
    font-family: 'Segoe UI', Arial, sans-serif;  
    color: #000000;  
    background-color: #ffffff;
```

```
    line-height: 1.6;
}
```

```
/* ----- Student Info Bar ----- */
```

```
.student-bar {
    background-color: #000000;
    color: #ffffff;
    text-align: center;
    padding: 6px 10px;
    font-size: 0.85rem;
    letter-spacing: 0.5px;
}
```

```
/* ----- Header ----- */
```

```
header {
    background: linear-gradient(135deg, #000000, #4d4d4d);
    color: #ffffff;
    text-align: center;
    padding: 40px 20px;
    text-shadow: 2px 2px 4px rgba(255, 255, 255, 0.3);
}
```

```
header h1 {
    margin: 0 0 10px 0;
    font-size: 2.4rem;
    letter-spacing: 1px;
}
```

```
/* ----- Navigation Bar ----- */
```

```
nav {
    background-color: #1a1a1a;
    padding: 12px 0;
    box-shadow: 0 4px 8px rgba(0, 0, 0, 0.3);
}
```

```
.navlinks {
    list-style: none;
    display: flex;
    justify-content: center;
    flex-wrap: wrap;
```

```
margin: 0;
padding: 0;
}
```

```
.navlinks li {
margin: 0 15px;
}
```

```
.navlinks a {
color: #ffffff;
text-decoration: none;
font-weight: bold;
padding: 6px 10px;
border-radius: 4px;
transition: all 0.3s ease;
}
```

```
/* Hover effect for navigation links */
.navlinks a:hover {
background-color: #ffffff;
color: #000000;
text-shadow: none;
}
```

```
/* ----- Main / Sections / Articles ----- */
main {
max-width: 960px;
margin: 0 auto;
padding: 20px;
}
```

```
section {
margin-bottom: 40px;
padding-top: 10px;
border-top: 2px solid #000000;
}
```

```
article h2 {
border-left: 6px solid #000000;
padding-left: 12px;
font-size: 1.6rem;
```

```
}
```

```
article h3 {  
    margin-top: 20px;  
    text-decoration: underline;  
}
```

```
/* ----- Tables ----- */
```

```
table {  
    width: 100%;  
    border-collapse: collapse;  
    margin-top: 15px;  
    box-shadow: 0 2px 6px rgba(0, 0, 0, 0.2);  
}
```

```
th, td {  
    border: 1px solid #000000;  
    padding: 10px;  
    text-align: left;  
}
```

```
th {  
    background-color: #000000;  
    color: #ffffff;  
}
```

```
tbody tr:nth-child(even) {  
    background-color: #f2f2f2;  
}
```

```
/* ----- Images / Figures ----- */
```

```
figure {  
    display: inline-block;  
    margin: 10px 20px 10px 0;  
    text-align: center;  
}
```

```
figure img {  
    border: 2px solid #000000;  
    border-radius: 8px;
```

```
    box-shadow: 3px 3px 8px rgba(0, 0, 0, 0.3);
}
```

```
figcaption {
    margin-top: 6px;
    font-style: italic;
    font-size: 0.9rem;
}
```

```
/* ----- Forms ----- */
```

```
form {
    background-color: #f7f7f7;
    border: 1px solid #000000;
    border-radius: 10px;
    padding: 25px;
    box-shadow: 4px 4px 10px rgba(0, 0, 0, 0.25);
    max-width: 500px;
}
```

```
form label {
    display: inline-block;
    width: 150px;
    font-weight: bold;
    margin-top: 10px;
}
```

```
form input[type="text"],
form input[type="email"],
form input[type="password"],
```

```
form input[type="date"],
form input[type="number"],
form select,
```

```
form textarea {
    border: 1px solid #000000;
    border-radius: 4px;
    padding: 6px;
    margin-top: 10px;
    width: 60%;
    font-family: inherit;
}
```

```
fieldset {  
  border: 1px solid #000000;  
  border-radius: 6px;  
  margin-top: 15px;  
}
```

```
legend {  
  font-weight: bold;  
  padding: 0 8px;  
}
```

```
form button {  
  background-color: #000000;  
  color: #ffffff;  
  border: none;  
  border-radius: 6px;  
  padding: 10px 20px;  
  margin: 15px 8px 0 0;  
  cursor: pointer;  
  font-weight: bold;  
  transition: all 0.3s ease;  
}
```

```
/* Hover effect for buttons */  
form button:hover {  
  background-color: #ffffff;  
  color: #000000;  
  border: 2px solid #000000;  
  box-shadow: 0 0 8px rgba(0, 0, 0, 0.5);  
}
```

```
/* ----- Canvas ----- */  
canvas {  
  border: 2px dashed #000000;  
  border-radius: 6px;  
  display: block;  
  margin-top: 10px;  
  background-color: #ffffff;  
}
```

```
/* ----- Content Cards ----- */
.card-container {
  display: flex;

  gap: 20px;
  flex-wrap: wrap;
  margin-top: 15px;
}

.card {
  flex: 1 1 250px;
  background: linear-gradient(160deg, #ffffff, #d9d9d9);
  border: 1px solid #000000;
  border-radius: 12px;
  padding: 20px;
  box-shadow: 4px 4px 12px rgba(0, 0, 0, 0.3);
  text-shadow: 1px 1px 1px rgba(0, 0, 0, 0.1);
  transition: transform 0.3s ease, box-shadow 0.3s ease;
}

/* Hover effect for cards */
.card:hover {
  transform: translateY(-6px);
  box-shadow: 6px 10px 18px rgba(0, 0, 0, 0.4);
}

.hover-btn {
  background-color: #000000;
  color: #ffffff;
  border: none;
  border-radius: 20px;
  padding: 8px 16px;
  margin-top: 10px;
  cursor: pointer;
  transition: all 0.3s ease;
}

.hover-btn:hover {
  background-color: #ffffff;
  color: #000000;
  border: 2px solid #000000;
}
```

```

}

/* ----- Footer ----- */
footer {
  background-color: #000000;
  color: #ffffff;
  text-align: center;
  padding: 18px;
  margin-top: 30px;
}

```

7. CODE EXPLANATION

Code	Explanation
* { box-sizing: border-box; }	Ensures padding and border are included within an element's declared width/height.
background: linear-gradient(135deg, ...)	Creates a smooth colour transition used for the header background.
text-shadow: 2px 2px 4px rgba(...)	Adds a soft shadow behind text to improve depth and readability.
.navlinks a:hover { ... }	Changes the background/text colour of a nav link only while the mouse hovers over it.
border-collapse: collapse;	Removes double borders between adjacent table cells for a clean grid look.
box-shadow: 4px 4px 10px rgba(...)	Adds a drop-shadow around the form and cards to create a raised, 3D appearance.
border-radius: 10px / 12px;	Rounds the corners of the form and content cards.
.card:hover { transform: translateY(-6px); }	Lifts a content card upward on hover, combined with a deeper shadow for a lift effect.
transition: all 0.3s ease;	Smoothly animates property changes (e.g. colour, shadow) over 0.3 seconds instead of an instant jump.

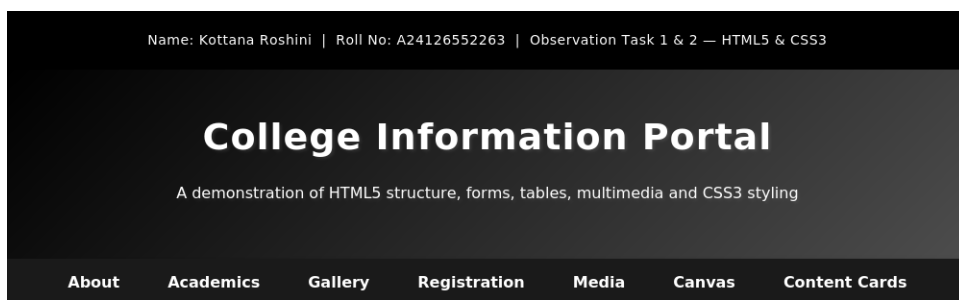
Code	Explanation
nth-child(even)	Selector used to shade alternate table rows for readability.

8. TEST CASES AND EXECUTION DATA

Test Case	Action	Expected Result	Actual Result	Status
TC-01	Open index.html with style.css linked	Page renders fully styled in black & white theme	Page rendered as styled	Pass
TC-02	Hover over navigation links	Link background turns white, text turns black	Hover effect worked correctly	Pass
TC-03	Hover over Submit/Reset buttons	Button colours invert with a visible border	Hover effect worked correctly	Pass
TC-04	Hover over a content card	Card lifts slightly and shadow deepens	Card lift effect worked correctly	Pass
TC-05	Inspect table rows	Alternating rows are shaded light grey	Alternating shading displayed correctly	Pass
TC-06	Inspect header section	Gradient background with shadowed heading text is visible	Gradient and text-shadow rendered correctly	Pass
TC-07	Resize browser window	Cards wrap responsively using flexbox	Cards wrapped correctly on smaller widths	Pass

9. OUTPUT

Output 1: Header Gradient, Text-Shadow and Navigation Hover Styling



About the College

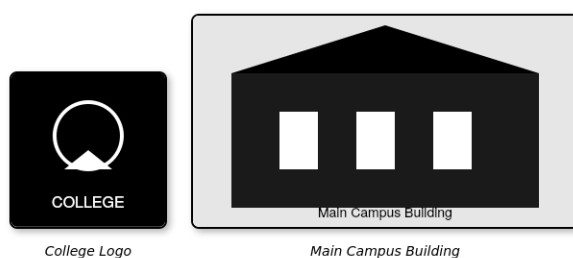
Our college is a premier institution dedicated to **academic excellence**, *innovation*, and holistic student development. Established with a vision to create future-ready engineers, the campus offers modern laboratories, experienced faculty, and a vibrant student life.

Formulae and notes often use special characters, for example water is written as H_2O and Einstein's famous equation is $E = mc^2$.

Ordered List — Admission Steps

1. Fill the online application form
2. Upload required documents
3. Pay the application fee

Output 2: Table Shading and Form Box-Shadow / Border-Radius



Student Registration Form

The image shows a registration form with a light grey background and rounded corners. It contains the following fields: "Name:" with a text input field containing "Enter your full name"; "Email:" with a text input field containing "you@example.com"; "Password:" with a text input field; and "Gender" with three radio button options: "Male", "Female", and "Other".

Output 3: Content Cards — Gradient Background, Box-Shadow and Hover Effect



My Styled Webpage — Content Cards

Card One Demonstrates border-radius, box-shadow and padding using a black & white theme. Learn More	Card Two Demonstrates gradient background and text-shadow effects. Learn More	Card Three Demonstrates hover effects and font styling across the page. Learn More
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10. RESULT

Thus, the College Information Portal web page was successfully styled using CSS3. Gradients, box-shadow, text-shadow, border-radius, alternating table shading and hover effects on navigation links, buttons and content cards were implemented in style.css and linked to index.html, and the styled output was verified using multiple test cases with the expected result obtained.